SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
... of Computer Science and Engineering

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Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	P. Greeshma	**Reg. No.:**	192424418
Section / Batch:	2024-AI&DS	**Date:**	28-07-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1

SECTION A – CONCEPT MAPPING

Explain how substitution method derives final complexity. Extend the map with Base Case and Induction Hypothesis nodes. Justify the need for proof in recurrence solving. Interpret how this map ensures correctness of derived complexity.

Code of Conduct

I P.Greeshma(192424418) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Greeshma P.Greeshma

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are wellstructured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%				
Linkages	25%				
Organization	20%				
Integration of Literature	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ______________	Signature: ______________	Signature: ______________
Date: ________________	Date: ________________	Date: ________________

SUBSTITUTION METHOD FOR RECURRENCE SOLVING

1. Guess The Complexity
Make An initial guess for the asymptotic complexity of $T(n)$

Example :- $T(n) = O(f(n))$

$O(n \log n)$?

2. Base Case
Verify that the recurrence satisfies the guessed bound for the smallest input (usually $n=1$) check $T(1)$

3. Induction Hypothesis
Assume the guessed bound is true for all smaller inputs

Assume :- $T(k) \le c \cdot f(k)$ for all $k<n$, where $c>0$ is a constant

Example
$T(n) = 2T(n/2)+n$

Guess : $O(n \log n)$

After apply substitution Method, $T(n) = \Theta(n \log n)$

4. Substitute Into Recurrence
Replace each recursive term (like $T(n/2)$, $T(n-1)$ etc. using the induction hypothesis

5. Simplify The Expression
perform algebraic simplification to obtain an upper bound.

Bound Holds
If $T(n) \le c \cdot f(n)$ for $n > n_0$, for some constant c and n_0, the gussed complexity is correct

6. Compare With The Gussed Bound
Check whether the resulting Expression satisfies the original gussed bound

Bound Does Not Hold
If the bound not satisfy revise the guess and repeat the process

7. Final Complexity
When both base case and induction step hold the reccurence is solved

$T(n) = \Theta(f(n))$